MATH 526 Lab 3 Fall 2006

Part I

In this part we perform Gaussian elimination without pivoting to obtain the **L-U** factorization of a 4×4 matrix.

Create the matrix

$$\mathbf{A} = \begin{bmatrix} 3 & 2 & -1 & 2 \\ -3 & -4 & 2 & 1 \\ 6 & 2 & -2 & 11 \\ -6 & -10 & 6 & 2 \end{bmatrix}$$

1. Initialize the matrix **L**.

>>L = eye(4)

2. Compute the multipliers and eliminate the entries of **A** in the first column. Observe that the matrix **L** is being created and the matrix **A** is being over-written in the process. Make sure you understand the commands.

>>L(2,1) = A(2,1)/A(1,1)

>>A(2,1:4) = A(2,1:4)–L(2,1)*A(1,1:4)

>>L(3,1)=A(3,1)/A(1,1)

>>A(3,1:4)=A(3,1:4)–L(3,1)*A(1,1:4)

>>L(4,1)=A(4,1)/A(1,1)

>>A(4,1:4)=A(4,1:4)–L(4,1)*A(1,1:4)

3. Compute the multipliers and eliminate the entries of **A** in the second column.

>>L(3,2)=A(3,2)/A(2,2)

>>A(3,2:4)=A(3,2:4)–L(3,2)*A(2,2:4)

>>L(4,2)=A(4,2)/A(2,2)

>>A(4,2:4)=A(4,2:4)–L(4,2)*A(2,2:4)

4. Compute the multipliers and eliminate the entries of **A** in the third column.

>>L(4,3)=A(4,3)/A(3,3)

>>A(4,3:4)=A(4,3:4)–L(4,3)*A(3,3:4)

5. The matrix **A** has been transformed to an upper triangular matrix. So we rename it **U**.

>>U=A

>>L

>>A=L*U

You should recover the original matrix **A**.

The commands above can be simplified by using **for** loops. Moreover, we can store the commands in an M-file.

Use an M-file window to create the following file, which starts with three comment lines:

```
% The function in this M-file computes the L-U factorization of a 4 × 4 matrix
% without pivoting. To use this function, create a 4 × 4 matrix A and type
% [L,U]=LU4(A).

function [L,U]=LU4(A)
L=eye(4);
for j=1:3
   for i=j+1:4
      L(i,j)=A(i,j)/A(j,j);
      A(i,j:4)=A(i,j:4)-L(i,j)*A(j,j:4);
   end
end
U=A;
```

Save this file as LU4.m

Now go back to your MATLAB window and type the following commands.

>>**clear L U**

>>**help LU4**

What appeared on the screen?

>>**[L,U]=LU4(A)**

The previously obtained **L** and **U** should reappear.

Clear all the variables before you work on Part II.